

2.4 Multimodal gradient competition

The phenomenon most structurally analogous to ours is gradient competition in multimodal learning. Wang, Tran, and Feiszli Wang et al. [2020] showed empirically that joint training of multimodal classification networks underperforms unimodal training because different modalities overfit and generalize at different rates. They proposed Gradient Blending to dynamically reweight modality-specific losses. Peng et al. [2022] identified “modality laziness” — a tendency for one modality to dominate training, leaving the other under-optimized — and proposed On-the-fly Gradient Modulation. These works are empirical; our contribution adapts and extends their framework to the unimodal *compositional* setting and provides a rigorous geometric foundation. The recent biomedical application of gradient blending to multimodal sarcoma prognosis Bozzo et al. [2024] confirms the mechanism operates in medical settings, but is restricted to explicit dual-modality (MRI + clinical) architectures, not the single-modality compositional pipelines we consider.

2.5 Spectral-spatial deep learning

Spectral-spatial deep learning has been an active area for over a decade, primarily in two communities: biomedical infrared spectroscopy and hyperspectral remote sensing. Berisha et al. [2019] established the benefit of combining spectral and spatial features for FTIR histology and observed that some tissue types could be classified from spatial features alone, with “minimal chemical information” required — a foreshadowing of the spatial shortcut phenomenon we diagnose. More recently, O’Leary et al. [2026] systematically compared spatial-spectral architectures for prostate cancer classification in infrared spectroscopy and reported the striking finding that compressing the spectral dimension to as few as 16 features had negligible effect on all models tested. The authors interpreted this as evidence that tissue classification requires only a limited spectral signature. Our Theorem 27 provides an alternative explanation: end-to-end training fails to extract the spectral signature even when it is present.

The conceptually nearest work in biomedical imaging is the recent characterization of shortcut learning in histopathology by Dawood and Minhas et al. [2026], who showed that AI models predicting molecular biomarkers from H&E images exploit spurious correlations with clinicopathological features rather than learning genuine biological signal. Our paper extends the shortcut-learning framework into a new modality (spectroscopic imaging) with a different mechanism (gradient competition driven by architectural asymmetry rather than confounder leakage).

2.6 Frozen and random features

Frozen random features have a long history in machine learning. Saxe et al. [2011] showed that convolutional networks with random (untrained) filters achieve competitive performance on standard benchmarks. Rahimi and Recht [2007] established the theoretical foundations of random features for kernel methods. More recently, Coil and Cheney [2025] showed that freezing layers acts as implicit regularization in overparametrized networks. These works establish that frozen random features are not an oddity but a principled choice. Our contribution is to identify the specific mechanism — gradient starvation under capacity asymmetry — under which freezing is not merely competitive but *strictly preferable* to joint training.

3 Setup and Preliminaries

We analyze a generic two-module compositional model. While our motivating application is infrared spectroscopic imaging for tissue classification, the framework applies to any pipeline

that first reduces a per-element spectral signal and then applies a high-capacity spatial model. We deliberately keep the notation domain-agnostic.

3.1 The compositional model

Let $X \in \mathbb{R}^{H \times W \times S}$ denote a hyperspectral input image, where H and W are spatial dimensions and S is the spectral dimension.¹ We write the per-pixel spectrum at location (h, w) as $x_{h,w} \in \mathbb{R}^S$.

The model consists of two modules applied sequentially:

1. A *spectral reduction* $f_{\theta} : \mathbb{R}^S \rightarrow \mathbb{R}^K$ applied independently to every pixel, producing a feature map $Z \in \mathbb{R}^{K \times H \times W}$ with $Z_{:,h,w} = f_{\theta}(x_{h,w})$.
2. A *spatial model* $g_{\phi} : \mathbb{R}^{K \times H \times W} \rightarrow \mathbb{R}^{N_{\text{cls}} \times H \times W}$ producing per-pixel class logits $\hat{y} = g_{\phi}(Z)$.

We denote the composed model $F_{\theta,\phi}(X) = g_{\phi}(f_{\theta}(X))$ and the parameter counts $C_f := \dim(\theta)$ and $C_g := \dim(\phi)$.

Assumption 1 (Linear spectral reduction). f_{θ} is linear in θ : there exists $W \in \mathbb{R}^{S \times K}$ with $\theta = \text{vec}(W)$ and $f_{\theta}(x_{h,w}) = W^{\top} x_{h,w}$.

Remark 2. Assumption 1 captures the canonical case used widely in spectroscopy pipelines, including the FTIR and QCL tissue classification work we draw motivation from. It is also the cleanest case in which the Hessian of $\mathcal{L}$ admits closed-form block analysis. We discuss extensions to shallow nonlinear f_{θ} in Section 9 and verify empirically in Section 7 that the theorem’s scaling persists when f_{θ} is replaced by a small MLP.

Assumption 3 (Spatial model regularity). g_{ϕ} is twice continuously differentiable in ϕ with L -Lipschitz gradients. For piecewise-smooth models (e.g. ReLU-type activations) this is relaxed: the gradient flow is interpreted in the almost-everywhere sense of Section 5, and no quantitative bound in this paper consumes second derivatives.

Assumption 4 (Bounded input second moment). Define the *mean empirical input Gram*

$$\Sigma_X := \frac{1}{N} \sum_{n=1}^N x_n x_n^{\top}, \quad N := N_{\text{data}} H W, \quad (1)$$

with x_n enumerating the per-pixel spectra of all images. We assume $\lambda_{\max}(\Sigma_X) \leq \Lambda_X < \infty$, with Λ_X uniform over the datasets under consideration. Every constant below is stated in terms of $\lambda_{\max}(\Sigma_X)$ and is therefore independent of the dataset size.

Remark 5 (Population motivation—not used in proofs). For a fixed finite dataset, $\lambda_{\max}(\Sigma_X)$ is finite automatically, and this empirical quantity is all the proofs consume; the assumption has content only along sequences of growing datasets. If the pixel spectra are modelled as sub-Gaussian draws from a population with second moment $\Sigma = \mathbb{E}[xx^{\top}]$ that are independent (or sufficiently weakly dependent) *across pixels*, then $\lambda_{\max}(\Sigma_X) \rightarrow \lambda_{\max}(\Sigma)$ and the uniform bound holds with Λ_X near $\lambda_{\max}(\Sigma)$. Marginal sub-Gaussianity alone does *not* suffice: neighbouring pixels of a tissue image are strongly correlated, and in the extreme case of perfectly duplicated pixels Σ_X stays bounded but never concentrates around Σ . This is why we assume the empirical bound directly rather than derive it from population conditions. Well-conditionedness of Σ (normalized inputs) is likewise invoked only in the discussion of classical remedies (Section 4.5), never in a proof.

¹Multi-channel inputs (e.g. raw absorbance together with first and second derivatives) are handled by concatenating channels along the spectral dimension; this changes only the input size and leaves the analysis unaffected.

Assumption 6 (Capacity gap). $C_g/C_f \rightarrow \infty$.

Assumption 6 is not used in any proof: it is the structural regime in which our theorems become *informative* rather than merely true, and we state it to delimit scope. Empirically it is satisfied with room to spare in modern spectral-spatial architectures: in our motivating example (Section 8, Table 2) the ratio is $C_g/C_f \approx 300$, roughly two orders of magnitude.

Assumption 7 (Bounded input Jacobian of the classifier). There exists a constant $\tilde{L} > 0$ depending on the depth of g_ϕ , the bottleneck dimension K , the number of classes N_{cls} , and the data distribution — but *not* on the characteristic width M or the total spatial parameter count C_g — such that

$$\sup_{\phi \in \Phi_{\text{reg}}, Z \in \mathcal{Z}_{\text{reg}}} \left\| \frac{\partial g_\phi(Z)}{\partial Z} \right\|_{\text{op}} \leq \tilde{L},$$

uniformly over the training-time parameter regime Φ_{reg} and the induced feature domain $\mathcal{Z}_{\text{reg}}$.

Remark 8 (Why Assumption 7 is separate from Assumption 3). Assumption 3 bounds the second derivatives of g_ϕ with respect to *parameters* ϕ (the standard L -smoothness condition of the optimization literature). Assumption 7 bounds the first derivative with respect to *input features* Z —a different object with a different constant. Parameter-side smoothness does not imply input-side Lipschitzness. Assumption 3’s role is to keep the gradient flow (7) well posed; the quantitative bounds of Theorems 14 and 27 use only Assumption 7.

What random-matrix theory does and does not supply here deserves care. For a random weight matrix whose *two dimensions grow proportionally*, variance-scaled initialization gives an operator norm that is $\Theta(1)$ in width: this is the classical largest-singular-value edge (Yin, Bai, and Krishnaiah Yin et al. [1988]; Bai and Yin Bai and Yin [1993]; non-asymptotically, Vershynin Vershynin [2018], Thm. 4.4.5; Marchenko–Pastur Marčenko and Pastur [1967] describes the corresponding bulk spectrum). It does *not* apply to the layer reading the bottleneck: a Kaiming-initialized expansion matrix $W_1 \in \mathbb{R}^{M \times K}$ with fixed K and entry variance σ_w^2/K has $\|W_1\|_{\text{op}} = \Theta(\sqrt{M/K})$, which *diverges* with width, so a per-layer product-of-operator-norms argument inherits this factor. The boundedness of the composed Jacobian $\partial g_\phi/\partial Z$ is instead a cancellation effect across consecutive variance-scaled layers: the $1/\sqrt{M}$ entry scaling of the *next* layer offsets the $\sqrt{M}$ growth, because the top directions of independent random factors are generically misaligned. We do not prove this in general—uniform boundedness is exactly what Assumption 7 postulates—but we prove it for a concrete two-layer ReLU head in Proposition 34 (supplement), where $\mathbb{E} \|\partial \hat{y}/\partial z\|_F^2 = \mathcal{O}(1)$ uniformly in width while the naive operator-norm product would give $\Theta(M/K)$. Under the maximal-update parameterization (μP), Yang and Hu Yang and Hu [2021] establish coordinate-wise $\Theta(1)$ stability of activations throughout training in the infinite-width limit; we take the corresponding operator-norm stability as a modeling assumption motivated by that result, not as a consequence of it. Under standard-parameterization training with weight decay $\lambda > 0$ and bounded-norm iterates, the per-layer product-of-operator-norms bound gives a *finite* $\tilde{L}$ over any training horizon—but not a width-uniform one, since per-layer norms at Kaiming scale grow like $\sqrt{M}$ and weight decay holds them near that scale rather than below it. Width-uniformity is therefore the modeling content of Assumption 7, supported at initialization by the composition-cancellation effect above (proved in expectation for the two-layer head) rather than derived from norm control. Softmax self-attention is Lipschitz only on bounded inputs Kim et al. [2021], so $\mathcal{Z}_{\text{reg}}$ should be taken compact whenever g_ϕ uses attention. Pre-attention LayerNorm (standard practice) supports this: it places each token on a sphere whose radius is set by the learned affine gains, so attention inputs stay bounded as long as those gains do, i.e. within the bounded parameter regime Φ_{reg} . We impose the compactness as part of the assumption rather than derive it.

3.2 Loss and gradients

We train with per-pixel cross-entropy, averaged over all N pixel positions of the dataset:

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \frac{1}{N} \sum_{n=1}^N \text{CE}(\hat{y}_n, y_n), \quad \hat{y}_n = F_{\boldsymbol{\theta}, \boldsymbol{\phi}}(X)_n, \quad (2)$$

where $y_n \in \{1, \dots, N_{\text{cls}}\}$ is the ground-truth class. Cross-entropy plays an essential role in Section 5: its gradient with respect to the logits $\hat{y}$ is the prediction residual $r_n = \text{softmax}(\hat{y}_n) - y_{n, \text{onehot}}$, which vanishes exponentially in the logit margin as predictions saturate—and hence, on separable data, polynomially in training time (Section 5). We discuss other loss functions sharing this vanishing-residual property in Section 9.

Normalization convention. Consistently with (2), we collect the per-pixel quantities into N -normalized stacked objects: the residual vector $r := N^{-1/2}(r_1, \dots, r_N)$, so that $\|r\|$ is the root-mean-square per-pixel residual, and the *logit Jacobian*

$$J_{\bullet} := N^{-1/2} \frac{\partial(\hat{y}_1, \dots, \hat{y}_N)}{\partial \bullet}, \quad \bullet \in \{\boldsymbol{\theta}, \boldsymbol{\phi}\}, \quad (3)$$

so that $\nabla_{\bullet} \mathcal{L} = J_{\bullet}^{\top} r$ exactly. Two Jacobian-like objects appear in the literature and we fix both here: $J_{\bullet}$ above (logit Jacobian), and the *residual Jacobian* $J_{\bullet}^{\text{res}} := \partial r / \partial \bullet = H_{\tau} J_{\bullet}$, where $H_{\tau} := \text{diag}(p) - pp^{\top}$ is the softmax Jacobian (blockwise over pixels). Since $0 \preceq H_{\tau} \preceq I$, any operator-norm bound on $J_{\bullet}$ applies a fortiori to $J_{\bullet}^{\text{res}}$. The intermediate factor $\partial Z / \partial \boldsymbol{\theta}$ carries the same $N^{-1/2}$ stacking, so that the chain rule composes as $J_{\boldsymbol{\theta}} = (\partial \hat{y} / \partial Z) \cdot (\partial Z / \partial \boldsymbol{\theta})$ with the unnormalized middle factor — which is the object Assumption 7 bounds. All Gram matrices, and Σ_X in (1), use this same $1/N$ normalization, so no constant in this paper depends on the dataset size.

The gradient of $\mathcal{L}$ with respect to the spatial parameters is computed by standard back-propagation. The gradient with respect to the spectral parameters $\boldsymbol{\theta}$ admits the chain-rule decomposition

$$\nabla_{\boldsymbol{\theta}} \mathcal{L} = \underbrace{r}_{\text{residual}} \cdot \underbrace{\frac{\partial \hat{y}}{\partial Z}}_{\text{spatial Jacobian}} \cdot \underbrace{\frac{\partial Z}{\partial \boldsymbol{\theta}}}_{\text{input}} = J_{\boldsymbol{\theta}}^{\top} r, \quad (4)$$

with all three factors in the normalization fixed above (the per-pixel loss gradients $\partial \mathcal{L} / \partial \hat{y}_n = r_n / N$ are absorbed into the $N^{-1/2}$ -scaled r and $\partial Z / \partial \boldsymbol{\theta}$). The softmax enters through the residual itself; the matrix H_{τ} appears explicitly only when differentiating r , as in J^{res} above and in the Gauss-Newton blocks below. The three factors play distinct roles in our analysis. The residual is small when predictions are confident. The spatial Jacobian depends on the current state of $\boldsymbol{\phi}$ and thus changes throughout training. The final factor is the input itself and is stationary. The product structure implies that if the residual decays to zero, the gradient to $\boldsymbol{\theta}$ vanishes regardless of the other two factors. This observation underlies Theorem 27.

Remark 9 (How much of this depends on cross-entropy?). The two theorems depend on the loss to different degrees. Theorem 14 is essentially loss-agnostic: the softmax Jacobian H_{τ} enters its proof only through the upper bound $H_{\tau} \preceq I$ and through the per-pixel weights $p_{\min}^{(n)}$ that the head hypothesis carries—both contribute constants and no M -dependence. Under squared error the inner matrix is the identity and the argument simplifies, giving the same width-linear disparity. Theorem 27 is more loss-specific: the polynomial residual envelope of Assumption 24 is the characteristic behavior of logistic-type losses on separable data (Soudry et al. Soudry et al. [2018]), whereas squared error on a well-conditioned target admits a faster (exponential) envelope. As noted after Assumption 24, a faster envelope only tightens the displacement bound; the mechanism—a fast module driving the residual down and starving a slow one through the shared factor—is not specific to cross-entropy, though the $\log(1/\delta)$ rate is.

3.3 The block Hessian

The joint Hessian of $\mathcal{L}$ decomposes into blocks aligned with the two parameter groups:

$$H = \nabla^2 \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \begin{pmatrix} H_{\boldsymbol{\theta}\boldsymbol{\theta}} & H_{\boldsymbol{\theta}\boldsymbol{\phi}} \\ H_{\boldsymbol{\phi}\boldsymbol{\theta}} & H_{\boldsymbol{\phi}\boldsymbol{\phi}} \end{pmatrix}, \quad (5)$$

with $H_{\boldsymbol{\theta}\boldsymbol{\theta}} \in \mathbb{R}^{C_f \times C_f}$, $H_{\boldsymbol{\phi}\boldsymbol{\phi}} \in \mathbb{R}^{C_g \times C_g}$, and cross blocks of appropriate sizes. Theorem 14 bounds the *ratio* of the two diagonal blocks' top eigenvalues (Definition 12) in terms of the spatial width, by bounding each block separately—not the condition number of H , which is a different and, here, degenerate quantity (Remark 13).

For curvature analysis we work with the generalized Gauss-Newton matrix, which for softmax cross-entropy is

$$G = J^\top H_\tau J, \quad J = [J_\boldsymbol{\theta}, J_\boldsymbol{\phi}], \quad (6)$$

with J the (N -normalized) logit Jacobian of (3) and H_τ the softmax Jacobian (Botev et al. Botev et al. [2017]). G is the positive semidefinite part of H that survives at interpolation, and $H = G$ exactly when the residual vanishes. Theorem 14 is stated for the diagonal blocks $G_{\boldsymbol{\theta}\boldsymbol{\theta}} = J_\boldsymbol{\theta}^\top H_\tau J_\boldsymbol{\theta}$ and $G_{\boldsymbol{\phi}\boldsymbol{\phi}} = J_\boldsymbol{\phi}^\top H_\tau J_\boldsymbol{\phi}$. The residual-Gram variant $J^{\text{res}\top} J^{\text{res}} = J^\top H_\tau^2 J$ obeys the same upper bounds unconditionally ($H_\tau^2 \preceq I$) and the same lower bound under a correspondingly weighted form of the head hypothesis (supplement Section A.1), so the results transfer across conventions with only the softmax weighting of the constants changing.

3.4 The Effective Gradient Ratio

We introduce a real-time diagnostic that tracks the asymmetry in gradient flow between the two modules.

Definition 10 (Effective Gradient Ratio). At training step t with parameters $(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)$, the Effective Gradient Ratio is

$$\text{EGR}(t) := \frac{\|\nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}{\|\nabla_{\boldsymbol{\phi}} \mathcal{L}(\boldsymbol{\theta}_t, \boldsymbol{\phi}_t)\|}.$$

Remark 11. EGR is a unitless quantity directly computable inside any training loop with negligible overhead. Section 6 bounds the decay of its numerator—the spectral gradient norm—by $B_T C / (1 + \mu t)$ under the hypotheses of Theorem 27, characterizes the ratio's trajectory empirically, and proposes a threshold criterion for detecting gradient starvation onset.

3.5 Notational conventions

Throughout, $\|\cdot\|$ denotes the Euclidean (ℓ_2) norm for vectors and the spectral norm for matrices; $\lambda_{\max}(\cdot)$ and $\lambda_{\min}(\cdot)$ denote the largest and smallest eigenvalues of a symmetric matrix; $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ its condition number. We write $A \succeq B$ to mean $A - B$ is positive semidefinite. The symbol $\Omega(\cdot)$ is used in the standard asymptotic sense.

Definition 12 (Module curvature disparity). For the Gauss-Newton matrix G of (6) with diagonal blocks $G_{\boldsymbol{\phi}\boldsymbol{\phi}}, G_{\boldsymbol{\theta}\boldsymbol{\theta}}$ indexed as in (5), the *module curvature disparity* is

$$\mathcal{D}_{\text{curv}}(G) := \frac{\lambda_{\max}(G_{\boldsymbol{\phi}\boldsymbol{\phi}})}{\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}})},$$

the ratio of the spatial block's top eigenvalue to the spectral block's, with the convention $\mathcal{D}_{\text{curv}} := +\infty$ when the spectral block vanishes ($\lambda_{\max}(G_{\boldsymbol{\theta}\boldsymbol{\theta}}) = 0$, a maximal disparity; Assumption 7 caps the spectral block from above only, so the degenerate case is not excluded, and lower bounds on $\mathcal{D}_{\text{curv}}$ are unaffected by it). Unless stated otherwise, $\mathcal{D}_{\text{curv}}$ is evaluated at random initialization $(\boldsymbol{\theta}_0, \boldsymbol{\phi}_0)$, which is where Theorem 14 bounds it and where Experiment 1.1 measures it.

Remark 13 (Why disparity, not condition number). Theorem 14 is a purely geometric statement about how curvature concentrates between the two modules as the spatial network widens. It is not a joint-Hessian condition-number bound: the joint Gauss-Newton G is rank-deficient (parameter directions orthogonal to the classifier’s range have exactly zero curvature), so $\kappa(G) = \lambda_{\max}(G)/\lambda_{\min}(G)$ is trivially infinite. The disparity $\mathcal{D}_{\text{curv}}$ is well-defined regardless of rank structure, uses only well-behaved operator-norm quantities, and is exactly the ratio measured in Experiment 1.1. Its growth with spatial capacity is the geometric mechanism that motivates the two-timescale dynamical analysis of Theorem 27, but the dynamical claim requires its own hypotheses and does not follow from $\mathcal{D}_{\text{curv}}$ alone.

4 Theorem 1: Capacity-Induced Module Curvature Disparity

This section establishes our first main result: the module curvature disparity $\mathcal{D}_{\text{curv}}$ (Definition 12) grows with the spatial module’s width. The result is the geometric foundation for the dynamical analysis of Section 5: it identifies a capacity-dependent mechanism capable of generating disparate module learning rates. The dynamical starvation claim itself is stated separately in Theorem 27, under its own explicit hypotheses.

4.1 Main result

Theorem 14 (Capacity-Induced Module Curvature Disparity). *Consider the compositional model $F_{\theta,\phi} = g_{\phi} \circ f_{\theta}$ with fixed-dimensional linear spectral module f_{θ} (Assumption 1) and a family of spatial networks $g_{\phi}^{(M)}$ indexed by characteristic width M (channels for a CNN, embedding dimension for a ViT, hidden width for an MLP). Under Assumptions 1, 4, and 7, together with the head hypothesis of Lemma 16 below (a dense classifier head over penultimate features of dimension $\Theta(M)$ that are correlated across inputs at initialization), there exist constants $c_1, c_2 > 0$ depending on the data distribution, the initialization scheme, the bottleneck dimension K , the input spectral dimension S , and the architecture class of g_{ϕ} (depth, activation, head dimensions)—but not on M or on the spatial parameter count C_g —such that*

$$\mathcal{D}_{\text{curv}}^{(M)} = \frac{\lambda_{\max}(G_{\phi\phi}^{(M)})}{\lambda_{\max}(G_{\theta\theta})} \geq c_1 \cdot M - c_2$$

in expectation over random Kaiming (variance-scaled Gaussian) initialization. Equivalently, for architecture families with $C_g = \Theta(M^2)$,

$$\mathcal{D}_{\text{curv}}^{(M)} \geq c'_1 \cdot \sqrt{C_g/C_f} - c_2.$$

Remark 15 (Interpretation). Theorem 14 states that as the spatial network widens, its top Gauss-Newton eigenvalue grows without bound while the spectral module’s top Gauss-Newton eigenvalue remains capped by a C_g -independent constant. Curvature increasingly concentrates in the downstream module. This is a purely geometric fact about the Hessian; the dynamical consequence—that joint training pins θ near its initialization—is stated separately in Theorem 27, under its own hypotheses. The disparity measures the downstream module’s *readiness to move*—the loss curvature its parameters command at initialization—not the usefulness of its features: it is present before any feature has been learned, and the width-linear spike can be exhibited even for inputs carrying no information (Remark 36, supplement). That the geometry favours the spatial module *independently* of feature quality is precisely what makes it a shortcut hazard.

The proof decomposes the disparity bound into separate bounds on the top eigenvalues of the two diagonal blocks. We state the supporting lemmas before presenting the proof.

4.2 Supporting lemmas

Lemma 16 (Spatial block scaling). *Let M denote the characteristic width of g_ϕ (channel count for a CNN, embedding dimension for a ViT, hidden layer width for an MLP), let $N_{\text{cls}} \geq 2$, and suppose g_ϕ ends in a dense classifier head over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram $S_h^p := N^{-1} \sum_n p_{\min}^{(n)} h_n h_n^\top$ (Equation (11)) satisfies*

$$\lambda_{\max}(S_h^p) \geq q_h M_h \quad \text{with probability } \geq 1 - \delta_h$$

over the random initialization—where $p_{\min}^{(n)}$ denotes the smallest softmax probability at pixel n —for constants $q_h > 0$ and $\delta_h < 1$ independent of M (the head hypothesis: the features carry a common direction of $\Theta(M)$ energy across inputs, on pixels whose initial predictions are non-degenerate. It is a cross-input correlation condition, not a per-vector norm condition, and is closely related to the quantity κ_2 carrying the width-linear term in Karakida et al.’s law—see the supplement, where a fully proved instance is also given). Then the spatial Gauss-Newton block of (6) satisfies

$$\lambda_{\max}(G_{\phi\phi}) = \Omega(M) \quad \begin{array}{l} \text{in expectation over random} \\ \text{Kaiming (variance-scaled} \\ \text{Gaussian) initialization.} \end{array}$$

The logit Gram $J_\phi^\top J_\phi \succeq G_{\phi\phi}$ inherits the bound unconditionally; the residual Gram $J_\phi^{\text{res}\top} J_\phi^{\text{res}}$ satisfies it under the analogous hypothesis on the p^2 -weighted Gram $S_h^{p^2}$ (supplement). For architecture families with $C_g = \Theta(M^2)$ (fixed depth with at least two consecutive width-proportional layers), equivalently $\lambda_{\max}(G_{\phi\phi}) = \Omega(\sqrt{C_g})$.

Remark 17. Lemma 16 is proved in the supplement (Section A.1) by an explicit witness: the rank-one head-weight perturbation $V = c u^\top$, where c is a unit class contrast and u the dominant direction of the feature ensemble. Evaluating the curvature along V gives the deterministic inequality $\lambda_{\max}(G_{\phi\phi}) \geq \lambda_{\max}(S_h^p)$, and the head hypothesis supplies $\lambda_{\max}(S_h^p) = \Omega(M)$. The width-linear law agrees with the mean-field analysis of Karakida et al. [2019]: their Theorem 4 gives $\lambda_{\max} = \Theta(M)$ with sharp constant for proportional-width fully connected networks under squared loss, and their Theorem 1 shows the mean eigenvalue simultaneously shrinks as $\mathcal{O}(1/M)$ —curvature concentrates in a vanishing fraction of directions as networks widen. The scaling variable is width, not parameter count: for architecture families with $C_g = \Theta(M^2)$, the equivalent parameter-count statement is $\lambda_{\max} = \Omega(\sqrt{C_g})$. The head hypothesis covers per-pixel dense heads on CNN and ViT modules as well; the sharpness of the width-linear rate is tested empirically in Section 7.

Lemma 18 (Spectral block operator-norm cap). *Under Assumptions 1 (linear per-pixel f_θ), 4 (bounded input second moment), and 7 (bounded input Jacobian $\|\partial g_\phi / \partial Z\|_{\text{op}} \leq \tilde{L}$ uniformly in C_g), the spectral logit Jacobian and Gauss-Newton block satisfy*

$$\lambda_{\max}(G_{\theta\theta}) \leq \|J_\theta\|_{\text{op}}^2 \leq \tilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) =: C_\theta,$$

where C_θ depends only on the classifier’s input Lipschitz constant $\tilde{L}$ and the top eigenvalue of the mean per-pixel input Gram Σ_X of (1)—not on the spatial width M , the spatial parameter count C_g , or the dataset size N . The same bound holds for the residual-Gram convention, since $H_\tau^2 \leq H_\tau \leq I$.

Remark 19. Lemma 18 follows from the chain-rule identity $J_\theta = J_Z \cdot (\partial Z / \partial \theta)$ for the logit Jacobian, combined with $\|J_Z\|_{\text{op}} \leq \tilde{L}$ (Assumption 7) and the exact form of $\partial Z / \partial \theta$ under linearity of f_θ (Assumption 1, giving the per-pixel factor $I_K \otimes x_n^\top$, whose N -normalized stacking has $\|\partial Z / \partial \theta\|_{\text{op}}^2 = \lambda_{\max}(\Sigma_X)$ exactly, by $I_K \otimes \Sigma_X$). Passing from the logit Jacobian to the

Gauss-Newton block or the residual Jacobian only inserts factors of $H_\tau \preceq I$ and can only decrease the bound. The bound is on the operator norm (equivalently, the top eigenvalue), which is well-behaved under rank-deficiency of the block—avoiding the smallest-eigenvalue and Kronecker-factorisation machinery discussed in Section A.1.

4.3 Proof of Theorem 14

Proof of Theorem 14. By Lemma 16, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G_{\phi\phi}^{(M)}) \geq c_\phi \cdot M$$

for a constant $c_\phi > 0$ independent of M (depending on the initialization scheme, the activation, the head dimensions, and the data distribution; supplement Section A.1). By Lemma 18,

$$\lambda_{\max}(G_{\theta\theta}) \leq \|J_\theta\|_{\text{op}}^2 \leq C_\theta$$

for a constant $C_\theta > 0$ independent of M . The cap is deterministic—it holds for every realization of the initialization—so it may be divided through inside the expectation:

$$\mathbb{E}[\mathcal{D}_{\text{curv}}^{(M)}] = \mathbb{E}\left[\frac{\lambda_{\max}(G_{\phi\phi}^{(M)})}{\lambda_{\max}(G_{\theta\theta})}\right] \geq \frac{\mathbb{E}[\lambda_{\max}(G_{\phi\phi}^{(M)})]}{C_\theta} \geq \frac{c_\phi \cdot M}{C_\theta} = c_1 \cdot M$$

with $c_1 := c_\phi/C_\theta$. The witness route delivers the bound with $c_2 = 0$; the subtracted constant is retained in the statement so that sharper asymptotic constants, valid only for large M , may be substituted. Under $C_g = \Theta(M^2)$ (families with at least two consecutive width-proportional layers), $M = \Theta(\sqrt{C_g})$ and the ratio form $c'_1 \cdot \sqrt{C_g}/C_f$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. $\square$

4.4 Implications

Theorem 14 establishes that curvature concentrates in the downstream spatial module as its width grows. We highlight two implications, and separate them explicitly from the dynamical consequences of Section 5.

Geometric mechanism, not a dynamical claim. $\mathcal{D}_{\text{curv}} \gg 1$ identifies a *potential* for disparate module learning rates: the spatial Gauss-Newton block admits directions of arbitrarily large curvature while the spectral block does not. Whether joint gradient descent actually produces disparate effective learning rates depends on an additional dynamical hypothesis (that the shortcut pathway fits the residual at rate μ ; Assumption 24), which Section 5 introduces and Experiment 1.2 tests. Theorem 14 does not, by itself, prove that θ is pinned to initialization. In particular, $\mathcal{D}_{\text{curv}}$ concerns the *top* of the spatial block’s spectrum, while the fitting rate μ of Section 5 is governed by how fast the residual can be driven down—a property of other parts of the spatial spectrum. The two are linked only under spectral-shape assumptions we do not make; the theorems are deliberately independent, and Experiments 1.1–1.2 test them separately.

The disparity is invariant to shared step-size tuning. The disparity bound depends only on architectural parameters, not on the choice of optimizer or the shared step-size schedule—so no choice of shared learning rate changes the *geometry*; whether any shared-step-size method escapes the *dynamical* consequence is a separate question we treat as a conjecture below. Adaptive methods (Adam, RMSProp) rescale per-coordinate gradient magnitudes but do not alter the underlying eigenvalue disparity between the two blocks. Two-timescale optimizers with explicitly different learning rates per module (Heusel et al. [2017]; Borkar 2008 Borkar [2008] Ch. 6) are the natural remedy at the optimizer level. We discuss this in Section 9.

4.5 Relation to classical ill-conditioning

Ill-conditioned optimization has classical causes with classical remedies, and it is natural to ask whether they apply here. They do not, because the disparity of Theorem 14 is driven by the *architecture*, not by the data or the parametrization of a single block.

Feature normalization. Scale asymmetry or collinearity among input features degrades the conditioning of Σ_X and is cured by standardizing inputs. The population picture behind Assumption 4 (Remark 5) already grants this cure: well-conditioned (e.g. normalized) inputs. The disparity persists regardless, because its growth is driven by the width-scaling of the spatial block (Lemma 16), which is untouched by any transformation of the input features.

Weight decay. Adding $\alpha \|w\|^2$ to the loss shifts the Hessian by $2\alpha I$, lifting every eigenvalue by 2α and famously repairing classical condition numbers. But the shift acts on *both* blocks equally: the disparity becomes $(\lambda_{\max}(G_{\phi\phi}) + 2\alpha)/(\lambda_{\max}(G_{\theta\theta}) + 2\alpha)$, which remains $\Theta(M)$ for any fixed α as M grows. Repairing the disparity would require $2\alpha \sim \lambda_{\max}(G_{\phi\phi})$, a regularization strength that would dominate the data term and prevent fitting. Consistent with this, Experiment 1.6 finds that Pezeshki’s Spectral Decoupling—an L_2 penalty on logits designed to combat saturation—does not reduce the joint-training collapse, while freezing does.

Adaptive optimizers. Adam and RMSProp precondition per-coordinate gradient magnitudes with a diagonal rescaling. A diagonal preconditioner can equalize per-parameter step sizes but does not restructure the eigenvalue disparity between the two blocks, which is a statement about the blocks’ spectra rather than their diagonals. Empirically, the canonical dynamics experiment (Experiment 1.2) is run under Adam and the joint-vs-frozen gap persists at every width; we treat the stronger claim—that no shared-learning-rate adaptive method can remove the pathology—as a conjecture rather than a theorem. Optimizers with explicitly different learning rates per module (two-timescale update rules; Heusel et al. [2017]) are the natural remedy at the optimizer level and correspond, in the limit, to our freezing prescription.

Remark 20 (Parameterization scope). $\mathcal{D}_{\text{curv}}$ is not invariant to per-module reparameterization, and Theorem 14 is a statement about the standard variance-scaled (Kaiming) parameterization—the one used in the pipelines we analyze and in all our experiments. Under alternative parameterizations that rescale layers with width, such as μP Yang and Hu [2021], both the initialization magnitudes and the effective per-layer learning rates change, and the width scaling of the blocks would need to be re-derived; we do not claim the disparity law transfers. Our two uses of μP elsewhere are limited to a different role: as one route to keeping the input-Jacobian bound (Assumption 7) uniform in width during training. Whether principled reparameterization can neutralize the disparity—rather than merely re-expressing it—is an open question adjacent to the per-module learning-rate remedies discussed in Section 9.

Remark 21 (Small eigenvalues: irrelevance vs. starvation). A parameter direction can carry vanishing curvature for two very different reasons: (a) the direction genuinely does not affect predictions on the task at hand (“irrelevance”), in which case zero gradient is the correct outcome and the parameter should not be learned; or (b) the direction *would* affect predictions if the optimizer could reach it, but the compositional geometry buries the gradient signal behind the bottleneck (“starvation”). The Gauss-Newton bounds of Theorem 14 alone cannot distinguish between these interpretations: both produce C_g -independent small curvature in the spectral block.

The synthetic experiments in Section 7 are designed to distinguish these hypotheses by construction. In Experiment 1.3 we generate data where each of the two modalities alone carries information sufficient to solve the task (single-modality baselines reach ≈ 0.53 on a binary classification, above the 0.50 chance floor). Joint training on the same data does not merely

under-use the spectral channel: it undershoots the weaker of the two single-modality baselines. The gap is modest in absolute terms on this synthetic problem—an important honest caveat—but its direction is inconsistent with interpretation (a), under which joint training would at worst match the stronger single-modality baseline.

For real spectroscopic data, distinguishing (a) from (b) requires demonstrating that a frozen pretrained spectral encoder outperforms one that has been jointly trained end-to-end on the same task. We report such experiments on FTIR and QCL tissue classification in Section 8. Our present study compares only two concrete mitigation strategies (freezing the spectral module, and Pezeshki’s Spectral Decoupling Pezeshki et al. [2021]); broader comparisons against modality-specific learning rates, gradient surgery, GradNorm-style loss balancing, and related interventions are important future work.

4.6 Empirical support

Experiment 1.1 in Section 7 measures $\mathcal{D}_{\text{curv}}^{(M)}$ directly across a range of spatial widths and confirms monotonic growth with M : the spatial top eigenvalue grows $74\times$ over the sweep while the spectral top eigenvalue stays flat, driving the disparity into the thousands. The measured log-log slope of λ_ϕ against C_g is 0.70 ± 0.05 , below the slope-1.0 prediction that applies to the tested architecture family (whose $C_g = \Theta(M)$); the deficit and its candidate causes are reported in Section 7. The qualitative prediction—unbounded spatial-block growth against a capped spectral block—is unambiguous in the data.

Remark 22 (Defense against linear triviality). A natural objection is that, since f_θ is linear, the analysis must reduce to principal component analysis (PCA) of the input. This is not the case. PCA finds directions of maximum input variance *without reference to labels*. Theorem 14 concerns the curvature of the supervised cross-entropy loss, which depends on the current state of g_ϕ through the Jacobian $\partial \hat{y} / \partial Z$. The pathology we identify is precisely that this supervised signal fails to reach θ , not that f_θ is incapable of representing useful features.

5 Theorem 2: Finite-Time Spectral Starvation

Theorem 14 is a geometric statement: as the spatial module widens, curvature concentrates in it and the spectral block’s top curvature stays capped. This section states the dynamical consequence we actually observe in training—the spectral module barely moves during the period in which the training loss is being fit—as a theorem with *explicit dynamical hypotheses*. We deliberately do not claim that the geometry of Theorem 14 alone implies the dynamics: the bridge from curvature disparity to rate separation is stated as a hypothesis, motivated by Theorem 14 and verified empirically in Section 7. Deriving it from the architecture is left as an open problem (Section 9).

Two classical facts shape the formulation. First, for unregularized cross-entropy on separable data there is *no finite minimizer*: the loss decays like $1/t$ while the parameters diverge along the max-margin direction (Soudry et al. Soudry et al. [2018]). A theorem asserting convergence to a finite stationary point would therefore be vacuous or false in the regime modern networks inhabit. Second, the starvation phenomenon of Pezeshki et al. Pezeshki et al. [2021] is fundamentally about which features receive learning signal *during training*, not about the $t \rightarrow \infty$ limit. Both point the same way: the honest statement is a *finite-time* bound on how far the spectral parameters can move before the shortcut pathway has fitted the labels.

5.1 Setting and hypotheses

We analyze the continuous-time limit of joint training with a shared learning rate, in which the parameters (θ_t, ϕ_t) evolve by gradient flow

$$\dot{\theta}(t) = -\nabla_{\theta} \mathcal{L}(\theta, \phi), \quad \dot{\phi}(t) = -\nabla_{\phi} \mathcal{L}(\theta, \phi). \quad (7)$$

All statements below hold for any absolutely continuous trajectory satisfying (7) for almost every t : under the smoothness of Assumption 3 the flow is classically well posed, and the almost-everywhere formulation additionally covers piecewise-smooth models such as ReLU networks, for which (7) is read as the corresponding subdifferential inclusion. A discrete-SGD extension (Robbins–Monro tracking, with gradient noise entering the constants) is a natural further step in the standard stochastic-approximation framework (Borkar Borkar [2008], Ch. 2); we do not develop it here (Remark 44, supplement).

Recall from the chain rule (4) that $\nabla_{\theta} \mathcal{L} = J_{\theta}^{\top} r$ where r is the softmax residual, so

$$\|\dot{\theta}(t)\| = \|J_{\theta}(t)^{\top} r(t)\| \leq \|J_{\theta}(t)\|_{\text{op}} \cdot \|r(t)\|. \quad (8)$$

Fix a horizon $T > 0$ and suppose the trajectory remains in the regime of Assumption 7, i.e. $\phi(t) \in \Phi_{\text{reg}}$ and $Z(t) \in \mathcal{Z}_{\text{reg}}$ for all $t \leq T$ (*trajectory containment*). Under Assumptions 1, 4, and 7, Lemma 18 then caps the first factor over that horizon:

$$B_T := \sup_{t \leq T} \|J_{\theta}(t)\|_{\text{op}} \leq \sqrt{C_{\theta}} = \tilde{L} \sqrt{\lambda_{\max}(\Sigma_X)}, \quad (9)$$

a constant independent of the spatial width M , the parameter count C_g , and the horizon T . (Even without containment, $B_T < \infty$ for every finite T by continuity of J_{θ} along the flow; containment is what makes the cap M -independent, which is the load-bearing property.) We write B for B_T when the horizon is clear. The remaining hypothesis concerns the second factor.

Remark 23 (Containment and the unregularized separable regime). Trajectory containment is a genuine restriction. In the unregularized separable regime that motivates the polynomial envelope below, parameter norms diverge—logarithmically for linear predictors (Soudry et al. Soudry et al. [2018]), and for homogeneous deep networks by the margin-maximization dynamics of Lyu and Li Lyu and Li [2020]—so a bound on $\|\partial g_{\phi}/\partial Z\|_{\text{op}}$ that is uniform over *all* t cannot be assumed. Two standard settings restore it: weight decay or norm constraints (bounded iterates; strictly, these modify the flow (7), and the theorem then applies to the modified flow, whose residual envelope may differ), and the maximal-update parameterization Yang and Hu [2021], which leaves the flow unchanged (activation-level width-uniform stability is established there; the operator-norm form we need is a motivated assumption, per the discussion after Assumption 7). In the pure max-margin limit, $\tilde{L}$ grows at most logarithmically in T , which turns the displacement bound of Theorem 27 into $\mathcal{O}(\varepsilon \log^2(1/\delta))$ —the same conclusion up to a log factor.

Assumption 24 (Shortcut residual decay). There exist constants $C \geq 1$ and $\mu > 0$ such that, along the joint trajectory (7),

$$\|r(t)\| \leq \frac{C}{1 + \mu t},$$

where $\|r(t)\|$ is the root-mean-square per-pixel residual of (3). We refer to μ as the *shortcut fitting rate*. The constants C and μ may depend on the initialization, the data, and the architecture—in particular μ is expected to grow with spatial capacity (Remark 25)—and are not assumed uniform over any family; the theorem holds for any trajectory admitting such an envelope.

Remark 25 (Why polynomial, and where μ comes from). The $1/t$ envelope is the correct cross-entropy-compatible form: for logistic-type losses on separable data the training loss—and hence the residual norm—decays polynomially, not exponentially (Soudry et al. Soudry et al. [2018]).

An exponential envelope, where valid (e.g. regularized or non-separable regimes), only strengthens the conclusion below. The hypothesis encodes the *shortcut* mechanism: the residual is driven down by the spatial module exploiting whatever signal is present in the feature map Z (positional and contextual structure), without requiring f_{θ} to specialize—the spatial-signal sufficiency discussed in Section 5.4. The rate μ is expected to *grow* with spatial capacity—wider networks fit faster, the familiar regularity of over-parameterized training; kernel-regime analyses tie fitting rates to spectral properties of the model’s kernel that improve with width (cf. the starvation analysis of Pezeshki et al. [2021]), though we use no such derivation here. We test the capacity dependence of the fitting speed empirically in Experiment 1.2. Deriving $\mu(C_g)$ from the architecture is an open problem we state explicitly at the end of Section 5.4; the theorem takes μ as given.

Definition 26 (Fitting time). For a target residual level $\delta \in (0, C)$, the fitting time is

$$T_{\delta} := \inf\{t \geq 0 : \|r(t)\| \leq \delta\}.$$

Under Assumption 24, $T_{\delta} \leq (C/\delta - 1)/\mu$.

5.2 Main result

Theorem 27 (Finite-Time Spectral Starvation). *Fix a horizon $T \geq 0$ and let $(\theta(t), \phi(t))$ be an absolutely continuous trajectory satisfying (7) for almost every $t \leq T$, contained in $\Phi_{\text{reg}} \times \mathcal{Z}_{\text{reg}}$ up to time T . Under Assumptions 1, 4, 7, and 24, with B_T as in (9), the trajectory satisfies*

$$\|\theta(T) - \theta_0\| \leq \frac{B_T C}{\mu} \log(1 + \mu T),$$

and, whenever $T \geq T_{\delta}$, at the fitting time T_{δ} ,

$$\|\theta(T_{\delta}) - \theta_0\| \leq \frac{B_T C}{\mu} \log\left(\frac{C}{\delta}\right).$$

In particular, writing $\varepsilon := B_T/\mu$ for the effective module-rate ratio, the spectral displacement by the time the training residual reaches level δ is $\mathcal{O}(\varepsilon \log(1/\delta))$.

Proof. By (8), the operator-norm cap (9), and Assumption 24, for almost every $t \leq T$,

$$\|\dot{\theta}(t)\| \leq B_T \cdot \|r(t)\| \leq \frac{B_T C}{1 + \mu t}.$$

Integrating,

$$\|\theta(T) - \theta_0\| \leq \int_0^T \|\dot{\theta}(t)\| dt \leq B_T C \int_0^T \frac{dt}{1 + \mu t} = \frac{B_T C}{\mu} \log(1 + \mu T).$$

For the second claim, Definition 26 gives $1 + \mu T_{\delta} \leq C/\delta$, so $\log(1 + \mu T_{\delta}) \leq \log(C/\delta)$. □

5.3 Interpretation

Theorem 27 is a statement about *training-time* behavior, which is what practice cares about: training runs end when the loss is fit (or by early stopping), not at $t = \infty$.

The starvation regime. The displacement bound is a product of factors with opposite capacity dependence. The factor B_T is capped independently of C_g (Lemma 18): no matter how large the spatial module, the spectral gradient cannot exceed $B_T \|r\|$. The fitting rate μ is *expected* to grow with spatial capacity (Remark 25; this is an empirical regularity we test, not a theorem): wider shortcut pathways drive the residual down faster. For families of architectures along which the envelope amplitude C stays bounded—as it does empirically, since $\|r(0)\|$ is $\Theta(1)$ at near-uniform initial predictions and the observed residuals decay from the start rather than plateauing—the displacement at the fitting time scales as $\varepsilon \log(1/\delta)$ with $\varepsilon = B_T/\mu \rightarrow 0$, and the spectral module is increasingly pinned to its initialization *for the entire useful duration of training*. We state the boundedness of C along the family as an explicit condition rather than a consequence: Assumption 24 deliberately does not make its constants uniform over architectures, so the capacity-asymptotic reading requires it. This is the capacity-worsens-starvation prediction tested in Experiment 1.2.

What we do not claim. We make no claim about the $t \rightarrow \infty$ limit. For unregularized cross-entropy on separable data no finite stationary point exists (Soudry et al. Soudry et al. [2018]), and θ may eventually move substantially over horizons exponentially longer than any practical training run ($T = \mathcal{O}(\varepsilon^{-q})$ still yields displacement $\mathcal{O}(\varepsilon \log(1/\varepsilon)) \rightarrow 0$, but larger horizons are unconstrained). The finite-time framing is not a weakness: it is the phenomenon. A spectral module that receives its learning signal only after the loss has been fitted receives it too late to shape which features the classifier uses.

The logarithm is benign. For any polynomial training horizon $T = \mathcal{O}(\varepsilon^{-q})$ with fixed q , and along families with bounded envelope amplitude C as above, the displacement is $\mathcal{O}(\varepsilon \log(1/\varepsilon))$, which vanishes as $\varepsilon \rightarrow 0$. The bound is thus meaningful across every practically reachable horizon, not merely at a single stopping time.

Relation to lazy training. Structurally, Theorem 27 is a *module-wise lazy training* bound: a bounded module Jacobian multiplied by an integrable residual envelope yields logarithmic parameter movement, in the spirit of Chizat, Oyallon, and Bach Chizat et al. [2019]. The differences are the driver and the scope. In the lazy-training literature, small movement follows from a scaling of the *model* (width or output scale) that makes the linearization accurate for *all* parameters; here, movement is small for *one module only*, it is driven by the interaction of two module-asymmetric quantities—the operator-norm cap of Lemma 18 and the shortcut fitting rate—and the other module is entirely free to travel. The phenomenon is a *selective* laziness induced by the compositional position of the spectral module, not a global linearization regime.

5.4 On the hypotheses

The operator-norm cap is proven, not assumed. The factor B_T is supplied by Lemma 18 under Assumptions 1 and 7. We are precise about how much of Section 4 enters this proof: only the cap—the same lemma that bounds $\mathcal{D}_{\text{curv}}$ ’s denominator. The width-linear growth of the spatial block (Lemma 16) appears nowhere in the derivation; its role is to motivate the capacity dependence of μ , which remains a hypothesis. Theorem 27 is a consequence of the cap plus the envelope, not of the disparity.

Residual decay encodes the shortcut. Assumption 24 is a statement about the *residual only*: the training loss gets fitted at rate μ along the joint trajectory. It does not mention θ , and in particular it does not assume the spectral module stays put. The shortcut interpretation—that the fitting is done by the spatial module exploiting structure already present in Z (positional

cues, contextual patterns)—is the mechanism that makes the assumption plausible at high spatial capacity, and it is satisfied in practice whenever the spatial model can overfit the training set, the norm for modern over-parameterized CNNs and ViTs. We test the envelope itself empirically in Experiment 1.2.

Remark 28 (Is the hypothesis circular?). A natural worry is that assuming fast residual decay comes close to assuming the conclusion. It does not, and the direction of the implication is the theorem’s content. The assumption constrains what the *loss* does; the theorem then caps how far θ can have travelled *on any trajectory consistent with that loss behavior*—including trajectories on which θ itself does part of the fitting. Nothing in the hypothesis privileges the spatial module: if a trajectory fits the residual at rate μ partly through spectral learning, the bound still applies and shows the total spectral displacement incurred in doing so is at most $(B_T C / \mu) \log(C/\delta)$. Read contrapositively, this is the sharpest form of the claim: *substantial spectral learning by the fitting time requires slow fitting*. A trajectory on which θ moves far by T_δ must have small μ —the loss must stay unfit for a long time—which is precisely what freezing-based pipelines engineer and what fast-fitting joint training precludes. (Over horizons far beyond the fitting time the logarithm can grow without bound; see “What we do not claim” below.) The hypothesis is also independently checkable: the envelope is a measurable property of a training run, verified or refuted by the run’s own residual curve, with no reference to θ .

Rate separation is motivated, not derived. Theorem 14 shows curvature disparity grows with capacity; it does not by itself prove that μ grows or that the trajectory satisfies Assumption 24. We keep the two theorems logically independent: Theorem 14 identifies a capacity-dependent geometric mechanism *capable* of generating disparate module learning rates; Theorem 27 characterizes the starvation dynamics *when* such a separation is present. Experiments 1.1–1.3 test the mechanism and the dynamics separately. Deriving Assumption 24—and in particular the growth of μ with C_g —from the architecture (e.g. via an NTK analysis of the spatial pathway at large width, adapting Pezeshki et al. [2021]) is, in our view, the most valuable missing theorem in this program: it would chain Theorems 14 and 27 into a single causal statement. We state it as an open problem rather than sketch it (Remark 44, supplement), and we test its content empirically: Experiment 1.2 measures the fitting speed as a function of spatial capacity.

6 The Effective Gradient Ratio Diagnostic

Theorems 14 and 27 together identify a training pathology that is invisible to *single-run* loss-curve monitoring. The failure is counterfactual: a jointly trained model converges to an apparently healthy solution—training loss down, validation metrics respectable, every curve shaped like a normal run—that is nonetheless worse than what the *same architecture* achieves with its spectral module frozen. Nothing in the run’s own curves flags this, because the deficit is visible only against a baseline the practitioner has not trained. One discovers it by training the frozen counterpart (twice the compute, and only if one thinks to try), or not at all.

This section introduces the *Effective Gradient Ratio* (EGR), a real-time diagnostic that detects gradient starvation as it occurs. EGR is a single scalar, computable inside any training loop at negligible cost, that converts the finite-time starvation prediction of Theorem 27 into a concrete observable.

6.1 Definition and basic properties

Recall from Definition 10 that at training step t ,

$$\text{EGR}(t) = \frac{\|\nabla_{\theta} \mathcal{L}(\theta_t, \phi_t)\|}{\|\nabla_{\phi} \mathcal{L}(\theta_t, \phi_t)\|}.$$

The numerator and denominator are computable directly from gradients accumulated during the standard backward pass. EGR is unitless and invariant to the choice of learning rate.

Initial value: a heuristic. A useful rule of thumb for the starting point of the diagnostic is

$$\text{EGR}(0) \sim \sqrt{C_f/C_g} \quad (\text{heuristic, not a theorem}). \quad (10)$$

The reasoning is a parameter-counting one: at variance-scaled initialization each module’s gradient has entries of comparable typical magnitude, so the squared gradient norms scale with the number of parameters carrying them, $\|\nabla_\phi \mathcal{L}\|^2 \sim C_g$ against $\|\nabla_\theta \mathcal{L}\|^2 \sim C_f$, and the square root comes from passing to norms. We state (10) as a heuristic rather than a proposition for three reasons: it exchanges the expectation of a ratio for the ratio of expectations without a concentration argument; the per-entry-magnitude premise is parametrization-dependent and would change under μP , which we cite elsewhere Yang and Hu [2021]; and the spectral module’s gradient additionally passes through the bottleneck, which the count ignores. Its role in this paper is limited to setting the scale for the *relative* threshold of Definition 32, which compares $\text{EGR}(t)$ against the measured $\text{EGR}(0)$ of the run at hand and so does not depend on (10) being sharp. The corresponding numerical value for our motivating architecture follows from the capacity ratio reported in Section 8.

6.2 Decay of the spectral gradient

Proposition 29 (Spectral gradient decay). *Under the hypotheses of Theorem 27—Assumptions 1, 4, 7, and 24, with the trajectory contained in $\Phi_{\text{reg}} \times \mathcal{Z}_{\text{reg}}$ up to a horizon T —we have, for every $t \leq T$,*

$$\|\nabla_\theta \mathcal{L}(t)\| \leq \frac{B_T C}{1 + \mu t}, \quad B_T \leq \sqrt{C_\theta} = \tilde{L} \sqrt{\lambda_{\max}(\Sigma_X)},$$

with B_T as in (9), C_θ the cap of Lemma 18, and C, μ the constants of Assumption 24.

Proof. Identical to (8): $\|\nabla_\theta \mathcal{L}\| = \|J_\theta^\top r\| \leq \|J_\theta\|_{\text{op}} \|r\| \leq B_T \|r\|$ for $t \leq T$, and Assumption 24 bounds $\|r(t)\|$. $\square$

Remark 30. The horizon restriction is not a technicality inherited for tidiness: Remark 23 notes that in the unregularized separable regime—the regime Assumption 24 models—a cap uniform over all $t \in [0, \infty)$ cannot be assumed. In practice T is the training run’s length, which is exactly the horizon the diagnostic operates over.

Remark 31 (From gradient decay to the EGR ratio). Proposition 29 bounds the *numerator* of the EGR. The denominator $\|\nabla_\phi \mathcal{L}\| = \|J_\phi^\top r\|$ is also proportional to the residual, so both quantities vanish together as the loss is fitted, and the behavior of their ratio is not determined by residual decay alone: it depends on the relative conditioning of the two Jacobians along the trajectory. We therefore treat the EGR trajectory as an empirical object, characterized in Section 6.5, and use Proposition 29 only for the absolute claim that the spectral module’s learning signal decays polynomially while the loss is being fitted.

6.3 Threshold criterion

The absolute decay of the spectral gradient (Proposition 29), combined with the empirical behavior of the ratio characterized in Section 6.5, motivates a practical criterion for detecting the onset of gradient starvation. We emphasize its status: the threshold rule is an empirically calibrated heuristic that the theory motivates but does not derive (Remark 31).

Definition 32 (Gradient starvation alert). We say the model is in a *gradient starvation regime* at step t if

$$\text{EGR}(t) < \tau \cdot \text{EGR}(0),$$

where $\tau \in (0, 1)$ is a user-supplied tolerance. Empirically, we recommend $\tau \in [0.05, 0.1]$, corresponding to a 10–20 $\times$ drop in EGR relative to initialization.

Remark 33. The threshold τ is the only hyperparameter of the diagnostic. The default $\tau = 0.1$ corresponds to a one-decade decay and is the value we use throughout Section 7.

Practical use. A practitioner logs $\text{EGR}(t)$ at each step. When the alert fires, two options are available:

1. **Freeze the spectral module.** This is the prescription most directly motivated by Theorem 27: removing θ from the optimization eliminates the timescale gap. We discuss this in Section 9.
2. **Inject a fixed residual baseline.** A softer intervention: replace $Z = f_\theta(X)$ with $Z = f_\theta(X) + h(X)$ for a fixed feature extractor h (e.g. PCA). The fixed baseline ensures the spatial model receives stable input even if f_θ continues to drift, breaking the shortcut without freezing.

We evaluate both interventions in Experiment 1.3.

6.4 Relationship to existing diagnostics

EGR is conceptually related to several existing notions but is distinct in its scope and intent.

GradNorm Chen et al. [2018]. GradNorm dynamically reweights losses in multi-task learning to equalize gradient magnitudes across tasks. EGR measures rather than intervenes, and applies to single-task compositional architectures rather than multi-task losses.

Modality-specific learning rates (MSLR). In multimodal learning, MSLR adjusts learning rates per modality. EGR diagnoses the underlying pathology that MSLR addresses but in a unimodal, two-stage architecture where the “modalities” are the spectral and spatial subspaces of a single input.

Loss curve monitoring. The conventional practice of monitoring training and validation loss fails to detect gradient starvation because both can decay even as θ stagnates. EGR is sensitive to the structural failure even when no global loss curve abnormality is present.

6.5 Empirical validation: EGR as a diagnostic

We validate EGR as a diagnostic of training pathology in two stages. First, we establish that EGR statistics correlate with generalisation across our full sweep. Second, we test whether the signal is genuine or merely a re-encoding of model capacity.

Pooled correlation (capacity confounded). Across $N = 24$ runs spanning four widths $\{16, 64, 256, 1024\}$ with six seeds each, EGR depth correlates strongly with final test accuracy: Pearson $r = -0.723$ ($p = 6.6 \times 10^{-5}$), Spearman $\rho = -0.808$ ($p = 1.8 \times 10^{-6}$), and with the train–test gap at $r = +0.741$ ($p = 3.5 \times 10^{-5}$). However, EGR depth is itself strongly correlated with log width ($r = +0.903$): roughly 77% of the pooled correlation with test accuracy is attributable to capacity rather than to EGR depth per se (partial r controlling for log width collapses to -0.169 , $p = 0.44$). The pooled correlation alone cannot establish a capacity-independent diagnostic.

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A Supplementary Material

This supplement provides the full proofs of Theorems 14 and 27, additional experiments referenced in the main text, and robustness checks.

A.1 Full proof of Theorem 14

We restate Theorem 14 for reference:

Under Assumptions 1, 4, and 7, together with the head hypothesis of Lemma 16, the module curvature disparity satisfies, in expectation over random Kaiming (variance-scaled Gaussian) initialization, $\mathcal{D}_{\text{curv}}^{(M)} \geq c_1 \cdot M - c_2$, and for architecture families with $C_g = \Theta(M^2)$, $\mathcal{D}_{\text{curv}} \geq c'_1 \sqrt{C_g/C_f} - c_2$, for constants depending on the data distribution, the initialization scheme, the bottleneck dimension K , the input spectral dimension S , and the architecture class of g_ϕ —but not on M or C_g .

The proof rests on two lemmas: (i) the spatial block’s top eigenvalue scales linearly with width M (Lemma 16, proved by an explicit witness direction in the head weights, with Karakida et al.’s mean-field law as the sharp asymptotic benchmark); (ii) the spectral block’s top eigenvalue is capped by a C_g -independent operator-norm bound (Lemma 18). We prove each lemma in turn, then combine.

A.1.1 Proof of Lemma 16

We show $\lambda_{\max}(G_{\phi\phi}) = \Omega(M)$ in expectation over random Kaiming (variance-scaled Gaussian) initialization, under the head hypothesis of Lemma 16: the classifier head of g_ϕ is a dense layer over penultimate features $h_n \in \mathbb{R}^{M_h}$ with $M_h = \Theta(M)$ whose softmax-weighted mean Gram has top eigenvalue $\Omega(M_h)$ at initialization. The proof has three steps: (1) context—Karakida et al.’s exact width-linear law for the top Fisher eigenvalue in their model class, which we use as the asymptotic benchmark but do *not* invoke directly; (2) an exact spectral lower bound on the softmax inner matrix at initialization; (3) an elementary explicit witness direction in the head weights that yields $\Omega(M)$ self-containedly. Throughout, Grams are normalized as empirical means over the $N := N_{\text{data}}HW$ pixel positions; the unnormalized (summed) Grams differ by the fixed factor N , which does not affect any statement in M .

Step 1: The width-linear law of Karakida et al. (context). Karakida, Akaho, and Amari Karakida et al. [2019] analyze fully connected networks of depth $L \geq 2$ with layer widths $M_\ell = \alpha_\ell M$, linear outputs, and variance-scaled Gaussian (Kaiming-type) initialization $W_{ij}^\ell \sim \mathcal{N}(0, \sigma_w^2/M_{\ell-1})$, $b_i^\ell \sim \mathcal{N}(0, \sigma_b^2)$. For the empirical Fisher information of the squared-loss model, $F = \frac{1}{T} \sum_{t=1}^T \sum_k \nabla f_k(x_t) \nabla f_k(x_t)^\top$ (the empirical-mean Gram of the output Jacobian), their Theorem 4 gives, in the limit $M \gg 1$,

$$\lambda_{\max}(F) = \alpha \left(\frac{T-1}{T} \kappa_2 + \frac{1}{T} \kappa_1 \right) M = \Theta(M),$$

where $\kappa_1, \kappa_2 = \Theta(1)$ are macroscopic constants computed from the layer-wise signal-propagation recursions and $\alpha = \sum_\ell \alpha_\ell \alpha_{\ell-1}$. Their Theorem 1 simultaneously shows the *mean* eigenvalue is $m_\lambda = C\kappa_1/M = \mathcal{O}(1/M)$: as the network widens, the FIM spectrum becomes extremely skewed—almost all directions flatten while the top eigenvalue grows linearly in width. We record two cautions. First, Theorem 4 fixes the *output* layer at $C = \mathcal{O}(1)$ but requires the input and hidden widths to scale proportionally with M ; a classifier reading a fixed bottleneck K violates the input-side requirement (its coefficient α degenerates there), so the theorem cannot be applied verbatim to our spatial modules. Second, both statements are asymptotic mean-field equalities with no finite- M rate. Steps 2–3 therefore prove the $\Omega(M)$ bound self-containedly;

Karakida’s law serves as the sharp benchmark within its model class, validated numerically in their Figure 1.

Step 2: The softmax inner matrix at initialization. Our loss is softmax cross-entropy, not squared loss, so the spatial Gauss–Newton block carries the softmax inner matrix

$$G_{\phi\phi} = \frac{1}{N} \sum_{n=1}^N J^{(n)\top} H_{\tau}^{(n)} J^{(n)}, \quad H_{\tau} := \text{diag}(p) - pp^{\top},$$

where $J^{(n)} := \partial \hat{y}^{(n)} / \partial \phi$ is the per-pixel logit Jacobian and $p = \text{softmax}(\hat{y})$. For any $v \in \mathbb{R}^{N_{\text{cls}}}$, the quadratic form is a variance: $v^{\top} H_{\tau} v = \sum_i p_i v_i^2 - (\sum_i p_i v_i)^2 = \text{Var}_{i \sim p}(v_i) \geq p_{\min} \sum_i (v_i - \bar{v}_p)^2 \geq p_{\min} \|P_{\perp} v\|^2$, where $p_{\min} := \min_i p_i$, $\bar{v}_p := \sum_i p_i v_i$, the last inequality uses that the unweighted mean minimizes the sum of squared deviations, and $P_{\perp} := I - \frac{1}{N_{\text{cls}}} \mathbf{1}\mathbf{1}^{\top}$ projects onto the mean-zero subspace $\mathbf{1}^{\perp}$. Hence

$$H_{\tau} \succeq p_{\min} P_{\perp}.$$

The bound is used per pixel in Step 3; we do not take a minimum over pixels, so no uniform-over-pixels lower bound on $p_{\min}$ is needed anywhere in the proof. (For orientation: at Kaiming initialization the logits are $\mathcal{O}(1)$ uniformly in M , so $p_{\min}^{(n)}$ is of order $1/N_{\text{cls}}$ for typical n .)

Step 3: An explicit witness direction. Write the classifier head of g_{ϕ} as $\hat{y}_n = Wh_n + b$ with penultimate features $h_n = h(x_n) \in \mathbb{R}^{M_h}$, $M_h = \Theta(M)$. A perturbation of the head weights is a matrix $V \in \mathbb{R}^{N_{\text{cls}} \times M_h}$; it is a unit vector in parameter space precisely when $\|V\|_F = 1$, under any vectorization convention, and the logit perturbation it induces at pixel n is Vh_n . Define the *softmax-weighted feature Gram*

$$S_h^p := \frac{1}{N} \sum_{n=1}^N p_{\min}^{(n)} h_n h_n^{\top}, \quad p_{\min}^{(n)} := \min_i p_i^{(n)}, \quad (11)$$

and let u be a top unit eigenvector of S_h^p . Fix a unit contrast $c \in \mathbb{R}^{N_{\text{cls}}}$ with $c \perp \mathbf{1}$ (possible since $N_{\text{cls}} \geq 2$); by Step 2, at every pixel

$$c^{\top} H_{\tau}^{(n)} c = \text{Var}_{i \sim p^{(n)}}(c_i) \geq p_{\min}^{(n)} \|P_{\perp} c\|^2 = p_{\min}^{(n)}.$$

Take the rank-one witness $V := cu^{\top}$, for which $\|V\|_F = \|c\| \|u\| = 1$ and $Vh_n = (h_n^{\top} u) c$. Then

$$\lambda_{\max}(G_{\phi\phi}) \geq \frac{1}{N} \sum_{n=1}^N (Vh_n)^{\top} H_{\tau}^{(n)} (Vh_n) = \frac{1}{N} \sum_{n=1}^N (h_n^{\top} u)^2 c^{\top} H_{\tau}^{(n)} c \geq u^{\top} S_h^p u = \lambda_{\max}(S_h^p). \quad (12)$$

Note what (12) is: a deterministic, hypothesis-free inequality—one identity and one application of Step 2—valid at every parameter value. All N pixels contribute, and no minimum over pixels is taken, so no union bound and no factor of N enters.

The head hypothesis of Lemma 16 is exactly $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1 - \delta_h$ over initialization, for $q_h > 0$ and $\delta_h < 1$ independent of M . On that event $\lambda_{\max}(G_{\phi\phi}) \geq q_h M_h = \Omega(M)$; since $\lambda_{\max} \geq 0$ always,

$$\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \geq (1 - \delta_h) q_h M_h = \Omega(M),$$

which is the claim.

On the head hypothesis, and its counterpart in Karakida et al. Write $S_h := N^{-1} \sum_n h_n h_n^\top$ for the unweighted mean feature Gram. The hypothesis $\lambda_{\max}(S_h^p) = \Omega(M_h)$ asks two things at once: that the penultimate features carry a common direction of $\Theta(M_h)$ energy across inputs ($\lambda_{\max}(S_h) = \Omega(M_h)$), and that the initial predictions are non-degenerate on the pixels supplying that energy. The second is mild at initialization—the logits are $\mathcal{O}(1)$ uniformly in M , since the head weights have variance $\Theta(1/M_h)$ against $\|h_n\|^2 = \mathcal{O}(M_h)$, so $p_{\min}^{(n)}$ is bounded below by a constant of order $1/N_{\text{cls}}$ on all but a small fraction of pixels, and S_h^p dominates that constant times the unweighted Gram restricted to those pixels. (Proposition 34 below turns this sketch into a proof for a concrete head.) The first is the substantive condition. It is implied by $\|\bar{h}\|^2 \geq qM_h$ on the *mean* feature vector $\bar{h} := N^{-1} \sum_n h_n$: for $\bar{u} = \bar{h} / \|\bar{h}\|$,

$$\lambda_{\max}(S_h) \geq \bar{u}^\top S_h \bar{u} = \frac{1}{\|\bar{h}\|^2} \cdot \frac{1}{N} \sum_n (h_n^\top \bar{h})^2 \geq \frac{1}{\|\bar{h}\|^2} \left(\frac{1}{N} \sum_n h_n^\top \bar{h} \right)^2 = \frac{\|\bar{h}\|^4}{\|\bar{h}\|^2} = \|\bar{h}\|^2,$$

the middle step being Jensen’s inequality (the mean of squares dominates the square of the mean). At variance-scaled initialization each entry of h_n has $\Theta(1)$ second moment (the signal-propagation recursions of Karakida et al. [2019], Eq. (9)), so the condition holds whenever the activations are correlated across inputs—in particular for any nonnegative or nonzero-mean activation (ReLU and its variants), and for odd activations whenever the bias variance $\sigma_b^2 > 0$ or the inputs are themselves correlated.

Note that per-entry second moments alone are *not* enough: $\|h_n\|^2 = \Theta(M_h)$ for every n gives only $\text{tr}(S_h) = \Theta(M_h)$, hence $\lambda_{\max}(S_h) \geq \Theta(M_h)/\min(N, M_h)$, which is weaker by a factor of N . Cross-input correlation, not per-vector norm, is what makes the top eigenvalue grow.

The same quantity carries the width-linear term in Karakida et al.’s Theorem 4, $\lambda_{\max} = \alpha(\frac{T-1}{T}\kappa_2 + \frac{1}{T}\kappa_1)M$: their κ_2 is built from the cross-input activation correlations $\hat{q}_{st}$. Their footnote notes that odd activations with $\sigma_b = 0$ give $\hat{q}_{st} = 0$ and $\kappa_2 = 0$, and is explicit that this exceptional case falls outside their analysis: “In such exceptional cases, we need to evaluate the lower order terms of s_λ and $\lambda_{\max}$ (outside the scope of this study).” We therefore attribute no finite- T claim to their theorem in the degenerate case. Our hypothesis has the parallel structure, and its degenerate behavior *can* be evaluated directly: for odd activations with $\sigma_b = 0$ and uncorrelated inputs the features lose their common direction, and the per-vector norms alone still give $\lambda_{\max}(S_h) \geq \text{tr}(S_h)/\min(N, M_h)$, so the hypothesis survives only with $q_h = \Theta(1/N)$: the width scaling remains, but the sample-size-independent leading constant is lost. For CNN and ViT heads we state the hypothesis as an explicit assumption rather than a theorem, and check it empirically in Section 7.

Proposition 34 (A fully proved instance). *Fix bottleneck features $z_1, \dots, z_N \in \mathbb{R}^K$ with $\|z_n\| \leq R$, and let the spatial module’s head be the two-layer ReLU network*

$$h_n = \text{ReLU}(W_1 z_n + b_1), \quad \hat{y}_n = W_2 h_n + b_2,$$

with Kaiming Gaussian initialization: entries of $W_1 \in \mathbb{R}^{M_h \times K}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/K)$, entries of $W_2 \in \mathbb{R}^{N_{\text{cls}} \times M_h}$ i.i.d. $\mathcal{N}(0, \sigma_w^2/M_h)$, biases i.i.d. $\mathcal{N}(0, \sigma_b^2)$ with $\sigma_b > 0$, and $N_{\text{cls}} \geq 2$. Let the spectral module be the linear per-pixel f_θ of Assumption 1 with inputs satisfying Assumption 4. Then there exist constants $q_h > 0$, $M_0 < \infty$, and $C_G < \infty$, depending only on $\sigma_w, \sigma_b, R, K, N_{\text{cls}}$, and $\text{tr}(\Sigma_X)$ —not on M_h or N —such that for all $M_h \geq M_0$, at initialization:

- (i) (head hypothesis) $\lambda_{\max}(S_h^p) \geq q_h M_h$ with probability at least $1/2$; consequently $\mathbb{E}[\lambda_{\max}(G_{\phi\phi})] \geq \frac{1}{2} q_h M_h$;
- (ii) (spectral-block cap, trace form) $\mathbb{E}[\lambda_{\max}(G_{\theta\theta})] \leq \mathbb{E}[\text{tr}(G_{\theta\theta})] \leq C_G := N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X)$, uniformly in M_h and N ;

(iii) (instance-level disparity law) with probability at least $3/8$, $\mathcal{D}_{\text{curv}} \geq (q_h/(8C_G)) M_h$; consequently $\mathbb{E}[\mathcal{D}_{\text{curv}}] \geq \frac{3}{64}(q_h/C_G) M_h = \Omega(M)$.

For this architecture, the disparity law of Theorem 14—not merely its hypotheses—is proved outright at initialization. (What remains assumption-level in general is the uniform-over-inputs, during-training form of Assumption 7: a per-input Jacobian cap at initialization follows from the computation in Step D below, but a supremum over all inputs by the natural union route would cost a $\sqrt{\log N}$ factor, and training-time persistence is exactly what Assumption 7 postulates. The trace-form cap (ii) bypasses both issues for the instance.)

Proof. Step A (mean feature vector). Write $g_{n,i} = w_i^\top z_n + b_{1,i}$ for the pre-activations, so $g_{n,i} \sim \mathcal{N}(0, s_n^2)$ with $s_n^2 = \sigma_w^2 \|z_n\|^2 / K + \sigma_b^2 \in [\sigma_b^2, s_{\max}^2]$, $s_{\max}^2 := \sigma_w^2 R^2 / K + \sigma_b^2$. Since $\mathbb{E} \text{ReLU}(\mathcal{N}(0, s^2)) = s/\sqrt{2\pi}$, each coordinate of the mean feature vector $\bar{h} = N^{-1} \sum_n h_n$ has expectation

$$m := \mathbb{E}[\bar{h}_i] = \frac{1}{N} \sum_n \frac{s_n}{\sqrt{2\pi}} \geq \frac{\sigma_b}{\sqrt{2\pi}} > 0.$$

To obtain constants free of the data and of N , we use throughout only the uniform floor $m \geq m_0 := \sigma_b/\sqrt{2\pi}$. The rows $(w_i, b_{1,i})$ are i.i.d. across i , so the coordinates $\bar{h}_i$ are i.i.d. with $\mathbb{E}[\bar{h}_i^2] \leq N^{-1} \sum_n \mathbb{E}[\text{ReLU}(g_{n,i})^2] \leq s_{\max}^2$ and (being averages of sub-Gaussian variables) finite fourth moment bounded by a constant $C_4(s_{\max})$. By Chebyshev's inequality applied to the i.i.d. sum $\|\bar{h}\|^2 = \sum_i \bar{h}_i^2$, whose mean is at least $m^2 M_h \geq m_0^2 M_h$,

$$\Pr\left[\|\bar{h}\|^2 < \frac{1}{2} m_0^2 M_h\right] \leq \frac{4 C_4(s_{\max})}{m_0^4 M_h} \xrightarrow{M_h \rightarrow \infty} 0.$$

On the complementary event E_A , the Jensen step above gives $\lambda_{\max}(S_h) \geq \|\bar{h}\|^2 \geq \frac{1}{2} m_0^2 M_h$, and moreover the witness direction $\bar{u} = \bar{h} / \|\bar{h}\|$ certifies $A := N^{-1} \sum_n (h_n^\top \bar{u})^2 \geq \frac{1}{2} m_0^2 M_h$.

Step B (softmax weights). Condition on $\mathcal{F}_1 = \sigma(W_1, b_1)$, which fixes the features h_n , the direction $\bar{u}$, and the energies $a_n := (h_n^\top \bar{u})^2$; the head randomness (W_2, b_2) is independent of $\mathcal{F}_1$. Call pixel n *tame* if $\|h_n\|^2 \leq 2s_{\max}^2 M_h$. For a tame pixel, the logits $\hat{y}_n | \mathcal{F}_1$ have i.i.d. Gaussian entries with variance $\sigma_w^2 \|h_n\|^2 / M_h + \sigma_b^2 \leq 2\sigma_w^2 s_{\max}^2 + \sigma_b^2 =: s_*^2$, so by a Gaussian tail bound on the logit range, there are constants $c_p(s_*, N_{\text{cls}}) > 0$ and $\delta_0 \leq 1/16$ with $\Pr[p_{\min}^{(n)} < c_p | \mathcal{F}_1] \leq \delta_0$ for every tame pixel. Untame pixels are exponentially rare: $\|h_n\|^2 / M_h$ is an average of M_h i.i.d. sub-exponential variables with mean $\leq s_{\max}^2/2$, so $\Pr[n \text{ untame}] \leq e^{-cM_h}$ (Bernstein's inequality; Vershynin Vershynin [2018], Thm. 2.8.1), and by Cauchy–Schwarz with $\mathbb{E} \|h_n\|^4 \leq 3s_{\max}^4 M_h^2$ (i.i.d. coordinates), the $\mathcal{F}_1$ -measurable untame energy $U := N^{-1} \sum_n a_n \mathbf{1}[n \text{ untame}]$ satisfies

$$\mathbb{E}[U] \leq \max_n \sqrt{\mathbb{E} \|h_n\|^4} \sqrt{\Pr[n \text{ untame}]} \leq \sqrt{3} s_{\max}^2 M_h e^{-cM_h/2} =: \varepsilon(M_h) M_h,$$

with $\varepsilon(M_h) \rightarrow 0$ exponentially (we used $a_n \leq \|h_n\|^2$, valid since $\bar{u}$ is a unit vector).

Step C (assembly). We control the two bad masses by separate arguments, respecting which randomness each depends on. The untame energy U is $\mathcal{F}_1$ -measurable; by (unconditional) Markov, the event $E_U := \{U \leq 16\varepsilon(M_h)M_h\}$ has $\Pr[E_U] \geq 15/16$, and for $M_h \geq M_0$ large enough that $16\varepsilon(M_h) \leq m_0^2/8$, on $E_A \cap E_U$ we have $U \leq \frac{1}{8} m_0^2 M_h \leq \frac{1}{4} A$. The p -degenerate mass over *tame* pixels, $P := N^{-1} \sum_{n \text{ tame}} a_n \mathbf{1}[p_{\min}^{(n)} < c_p]$, is (W_2, b_2) -random given $\mathcal{F}_1$, with $\mathbb{E}[P | \mathcal{F}_1] \leq \delta_0 A$ by Step B; conditional Markov with $\delta_0 \leq 1/16$ gives $\Pr[P > \frac{1}{4} A | \mathcal{F}_1] \leq 4\delta_0 \leq 1/4$. On the intersection of the three events—probability at least $1 - 4C_4/(m_0^4 M_h) - 1/16 - 1/4 > 1/2$ for $M_h \geq M_0$ —the total bad mass is $P + U \leq \frac{1}{2} A$, and

$$\lambda_{\max}(S_h^p) \geq \bar{u}^\top S_h^p \bar{u} \geq c_p (A - P - U) \geq \frac{c_p}{2} A \geq \frac{c_p m_0^2}{4} M_h.$$

Take $q_h := c_p m_0^2/4 = c_p \sigma_b^2/(8\pi)$, manifestly free of the data and of N . The in-expectation clause follows from $\lambda_{\max} \geq 0$ and (12). This proves (i).

Step D (spectral-block cap, trace form). At any input z , the head’s input Jacobian is $\partial \hat{y}/\partial z = W_2 D_z W_1$ with $D_z = \text{diag}(\mathbf{1}[W_1 z + b_1 > 0])$. Conditioning on (W_1, b_1) (which fixes D_z) and using that the rows of W_2 are i.i.d. $\mathcal{N}(0, (\sigma_w^2/M_h)I)$,

$$\mathbb{E} \|W_2 D_z W_1\|_F^2 = N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E} \|D_z W_1\|_F^2 \leq N_{\text{cls}} \cdot \frac{\sigma_w^2}{M_h} \cdot \mathbb{E} \|W_1\|_F^2 = N_{\text{cls}} \sigma_w^4,$$

using $\mathbb{E} \|W_1\|_F^2 = M_h K \cdot \sigma_w^2/K = M_h \sigma_w^2$ —uniformly over inputs z . (The naive product bound $\|W_2\|_{\text{op}} \|W_1\|_{\text{op}}$ would grow like $\sqrt{M_h/K}$; the composition is $O(1)$ because the top directions of the two random factors are generically misaligned—which the Frobenius computation captures without any alignment analysis.) Now bound the spectral block. With $J_{\theta}^{(n)} = (W_2 D_{z_n} W_1)(I_K \otimes x_n^\top)$ the per-pixel spectral logit Jacobian (Lemma 18, Step 3) and $H_\tau \preceq I$,

$$\mathbb{E} \text{tr}(G_{\theta\theta}) \leq \frac{1}{N} \sum_{n=1}^N \mathbb{E} \|J_{\theta}^{(n)}\|_F^2 \leq \frac{1}{N} \sum_{n=1}^N \mathbb{E} \|W_2 D_{z_n} W_1\|_F^2 \cdot \|x_n\|^2 \leq N_{\text{cls}} \sigma_w^4 \cdot \frac{1}{N} \sum_n \|x_n\|^2 = N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X),$$

using $\|AB\|_F \leq \|A\|_F \|B\|_{\text{op}}$ and $\|I_K \otimes x_n^\top\|_{\text{op}} = \|x_n\|$. Since $\lambda_{\max} \leq \text{tr}$ for positive semidefinite matrices, (ii) follows with $C_G = N_{\text{cls}} \sigma_w^4 \text{tr}(\Sigma_X)$, free of M_h and N .

Step E (combining). Let E_1 be the event of (i) ($\Pr \geq 1/2$) and $E_2 := \{\lambda_{\max}(G_{\theta\theta}) \leq 8C_G\}$, which by (ii) and Markov has $\Pr[E_2] \geq 7/8$. On $E_1 \cap E_2$ (probability at least $1 - 1/2 - 1/8 = 3/8$), the witness bound (12) gives

$$\mathcal{D}_{\text{curv}} = \frac{\lambda_{\max}(G_{\phi\phi})}{\lambda_{\max}(G_{\theta\theta})} \geq \frac{q_h M_h}{8C_G},$$

and since $\mathcal{D}_{\text{curv}} \geq 0$, $\mathbb{E}[\mathcal{D}_{\text{curv}}] \geq \frac{3}{8} \cdot q_h M_h / (8C_G) = \frac{3}{64} (q_h / C_G) M_h$. This proves (iii). $\square$

Remark 35. Proposition 34 is deliberately minimal: one hidden layer, ReLU, bias variance $\sigma_b^2 > 0$ (which guarantees activation correlation across inputs; cf. the κ_2 discussion above). Its role is to certify that Theorem 14’s two structural hypotheses are *theorems*, not merely assumptions, for at least one concrete member of each side of the pipeline—so the disparity law has a fully verified instance. For deeper heads and attention-based modules the hypotheses revert to assumption status, argued by signal propagation and checked empirically in Section 7.

Remark 36 (What the instance shows—and what it does not). The mechanism of Proposition 34 deserves a candid reading. The width-linear spike is generated by the *mean feature direction*: with $\sigma_b > 0$, every hidden unit has positive mean activation, so the features of all inputs share a common direction of $\Theta(M_h)$ energy. This requires no information in the features at all. Take every bottleneck feature $z_n = 0$, so that all inputs are indistinguishable: the features $h_n = \text{ReLU}(b_1)$ are identical across pixels, carry no sample-specific structure whatsoever, and the proposition applies verbatim. Width-linear downstream curvature is therefore *not* evidence that the spatial module has found, or will find, useful structure: $\mathcal{D}_{\text{curv}}$ measures the downstream module’s readiness to move—the loss curvature its parameters command at initialization—not the value of what it computes. This is the spatial-side counterpart of Remark 21: there, small spectral curvature cannot distinguish irrelevance from starvation; here, large spatial curvature cannot distinguish signal from capacity. It is also exactly why the geometry is a shortcut *hazard*: the capacity advantage is in place before the data has spoken, so whatever the spatial module can most easily fit—signal or shortcut—it is poised to fit first.

Remark 37 (Smoothness scope). ReLU is not twice differentiable, but no smoothness enters Proposition 34: the Gauss–Newton blocks involve only first-order Jacobians, which at Gaussian initialization exist almost surely (every preactivation is a.s. nonzero). Assumption 3 plays no

role here—its function is the well-posedness of the gradient flow in Section 5, whose statements are formulated for absolutely continuous trajectories satisfying the flow almost everywhere and therefore accommodate piecewise-smooth models.

The other two conventions. The bound just proved is for the Gauss-Newton block $G_{\phi\phi} = J_\phi^\top H_\tau J_\phi$ of (6), which is the object appearing in Definition 12. Two neighbouring matrices inherit it. For the *logit* Gram, $H_\tau \preceq I$ gives $J_\phi^\top J_\phi \succeq G_{\phi\phi}$, so $\lambda_{\max}(J_\phi^\top J_\phi) \geq \lambda_{\max}(G_{\phi\phi}) = \Omega(M)$ immediately. For the *residual* Gram $J_\phi^{\text{res}\top} J_\phi^{\text{res}} = J_\phi^\top H_\tau^2 J_\phi$, note that H_τ is symmetric with $H_\tau \mathbf{1} = 0$, so $\mathbf{1}^\perp$ is invariant and, by Step 2, $H_\tau^{(n)} \succeq p_{\min}^{(n)} I$ there; squaring the eigenvalues on that subspace gives $(H_\tau^{(n)})^2 \succeq (p_{\min}^{(n)})^2 P_\perp$, so the identical witness argument runs with the weights $(p_{\min}^{(n)})^2$ in place of $p_{\min}^{(n)}$ and yields $\lambda_{\max}(J_\phi^{\text{res}\top} J_\phi^{\text{res}}) \geq \lambda_{\max}(S_h^{p^2}) = \Omega(M)$ under the corresponding head hypothesis. All three conventions therefore satisfy Lemma 16, differing only in the softmax weighting carried by the feature Gram. This completes the proof. $\square$

Remark 38 (Scope, and the role of the mean-field law). The witness argument uses only: (a) a dense classifier head over penultimate features of dimension $\Theta(M)$; (b) the head hypothesis $\lambda_{\max}(S_h^p) \geq q_h M_h$ at initialization—a *cross-input correlation* condition, which variance-scaled initialization does not by itself deliver (see the counterexample of per-vector norms above) but which holds for the standard activation choices discussed there; and (c) softmax cross-entropy with $N_{\text{cls}} \geq 2$. Given (b), the inequality $\lambda_{\max}(G_{\phi\phi}) \geq \lambda_{\max}(S_h^p)$ is deterministic and architecture-agnostic, so the argument applies to per-pixel dense heads on MLP, CNN (channel count M), and ViT (embedding dimension M) spatial modules alike. For fully connected modules (b) follows from standard signal propagation at Kaiming initialization; for CNN and ViT heads we adopt it as an explicit assumption checked empirically. No i.i.d. Gaussian input assumption is required. Karakida et al.’s Theorem 4 plays the role of the sharp asymptotic benchmark: within its model class (fully connected, input and hidden widths proportional to M , output fixed at $C = \mathcal{O}(1)$, squared loss), it upgrades the witness lower bound to an exact width-linear law with explicit constant, and their Theorem 1 supplies the complementary fact that the *mean* eigenvalue shrinks as $\mathcal{O}(1/M)$, so curvature concentrates in a vanishing fraction of directions as the module widens. The loss transfer from squared loss to cross-entropy is handled exactly at initialization by Step 2, at the cost of the constant $p_{\min}$.

Remark 39 (Empirical status). Karakida et al.’s own numerical validation (their Figure 1, right column) confirms $\lambda_{\max}$ growing linearly in M , matching the exact constant of their Theorem 4, for tanh, ReLU, and linear fully connected networks. Our Experiment 1.1 tests the law on the SpatialMLP family, whose parameter count is *linear* in width ($C_g = \Theta(M)$, since input and output dimensions are fixed), so the width-linear law predicts a log-log slope of 1.0 against C_g as well as against M . The measured slope is 0.70 ± 0.05 : the qualitative claim of Lemma 16— λ_ϕ grows without bound while λ_θ stays capped—is confirmed unambiguously, but the measured exponent is sublinear, short of the width-linear asymptote. Candidate causes (finite width, the loss-Hessian versus Gauss–Newton measurement gap, correlated non-Gaussian features) are discussed in Section 7. We report the discrepancy rather than explain it away.

A.1.2 Proof of Lemma 18

The proof factors the spectral Jacobian via the chain rule and bounds each factor’s operator norm using Assumptions 1, 4, and 7. We bound the *logit* Jacobian $J_\theta = \partial \hat{y} / \partial \theta$ of (3); the Gauss-Newton block and the residual Jacobian then inherit the bound, since both insert factors of H_τ with $0 \preceq H_\tau \preceq I$ (Step 4).

Step 1: Chain-rule factorisation. With $Z = f_{\theta}(X)$ and $\hat{y} = g_{\phi}(Z)$, the chain rule gives, in the normalization of (3) (the $N^{-1/2}$ riding on the right-hand factor, as fixed in Section 3),

$$J_{\theta} = J_Z \cdot \frac{\partial Z}{\partial \theta}, \quad J_Z := \partial g_{\phi} / \partial Z,$$

where J_Z is unnormalized—the object Assumption 7 bounds. Since g_{ϕ} acts on one image at a time, the dataset-stacked J_Z is block diagonal across images, so its operator norm is the maximum over per-image blocks and Assumption 7 applies to the stack with the same constant $\tilde{L}$. (For the residual map $r(\theta, \phi) = \tau(g_{\phi}(Z))$ with $\tau(\hat{y}) = \text{softmax}(\hat{y}) - y_{\text{onehot}}$, the same computation carries the extra left factor $J_{\tau} := \partial \tau / \partial \hat{y} = \text{diag}(p) - pp^{\top} = H_{\tau}$, the softmax Jacobian, which satisfies $\|H_{\tau}\|_{\text{op}} \leq 1$; see Botev, Ritter, and Barber Botev et al. [2017].)

Step 2: Operator-norm inequality. Sub-multiplicativity of the operator norm gives

$$\|J_{\theta}\|_{\text{op}} \leq \|J_Z\|_{\text{op}} \cdot \left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}} \leq \tilde{L} \cdot \left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}},$$

using Assumption 7 on the first factor.

Step 3: The input-Jacobian factor, exactly. Under Assumption 1, $Z_n = W^{\top} x_n$ with $\theta = \text{vec}(W)$, $W \in \mathbb{R}^{S \times K}$. The per-pixel Jacobian is $\partial Z_n / \partial \text{vec}(W) = I_K \otimes x_n^{\top}$ (Magnus and Neudecker Magnus and Neudecker [1988], Ch. 9). Stacking over the N pixel positions with the $N^{-1/2}$ normalization of (3) gives $\partial Z / \partial \theta \in \mathbb{R}^{N \times SK}$ with

$$\left(\frac{\partial Z}{\partial \theta} \right)^{\top} \frac{\partial Z}{\partial \theta} = \frac{1}{N} \sum_{n=1}^N (I_K \otimes x_n) (I_K \otimes x_n^{\top}) = I_K \otimes \Sigma_X,$$

hence, *exactly*,

$$\left\| \frac{\partial Z}{\partial \theta} \right\|_{\text{op}}^2 = \lambda_{\max}(I_K \otimes \Sigma_X) = \lambda_{\max}(\Sigma_X),$$

with Σ_X the mean per-pixel input Gram of (1); Assumption 4 bounds $\lambda_{\max}(\Sigma_X) \leq \Lambda_X$ uniformly in the dataset size N . Note that no factor of K appears: the Kronecker structure replicates the same spectrum K times rather than accumulating it.

Step 4: Assembling. Combining Steps 2 and 3,

$$\|J_{\theta}\|_{\text{op}}^2 \leq \tilde{L}^2 \cdot \lambda_{\max}(\Sigma_X) =: C_{\theta}.$$

For the Gauss-Newton block, $G_{\theta\theta} = J_{\theta}^{\top} H_{\tau} J_{\theta} \preceq J_{\theta}^{\top} J_{\theta}$ since $H_{\tau} \preceq I$, so $\lambda_{\max}(G_{\theta\theta}) \leq C_{\theta}$; for the residual Jacobian, $\|H_{\tau} J_{\theta}\|_{\text{op}} \leq \|J_{\theta}\|_{\text{op}}$ gives the same cap. The constant C_{θ} depends on the classifier’s input Lipschitz constant $\tilde{L}$ (which may grow with the depth of g_{ϕ} but not with its width M ; see the discussion of Assumption 7) and the top eigenvalue of the mean input Gram $\lambda_{\max}(\Sigma_X)$. It does not depend on M , on the spatial parameter count C_g , or on the dataset size N . This completes the proof. $\square$

Remark 40 (Why an operator-norm cap suffices). An alternative route via the smallest positive eigenvalue $\lambda_{\min}^+(J_{\theta}^{\top} J_{\theta})$ requires either an exact Kronecker factorisation (valid only for per-pixel classifiers) or a trace-over-rank argument via the softmax residual dimension—both of which need additional structural assumptions on g_{ϕ} . The operator-norm bound above is significantly simpler, applies to general nonlinear spatial classifiers (CNN, ViT), and is exactly what Theorem 14 needs: a C_g -independent cap on the spectral block’s *top* eigenvalue, so that $\mathcal{D}_{\text{curv}} = \lambda_{\max}(G_{\phi\phi}) / \lambda_{\max}(G_{\theta\theta}) \geq \Omega(M) / C_{\theta}$.

Remark 41 (Softmax Jacobian bound). The bound $\|\text{diag}(p) - pp^\top\|_{\text{op}} \leq 1$ is standard: $\text{diag}(p) - pp^\top$ is the covariance matrix of the categorical distribution with parameter p (equivalently, the Jacobian of the softmax map), hence positive semi-definite with $\lambda_{\max} \leq \max_i p_i \leq 1$. See Botev, Ritter, and Barber Botev et al. [2017] for its role in Gauss–Newton computations.

A.1.3 Combining the lemmas: full proof of Theorem 14

Assemble Lemmas 16 and 18 to complete the proof. By Lemma 16, in expectation over random Kaiming (variance-scaled Gaussian) initialization,

$$\lambda_{\max}(G_{\phi\phi}^{(M)}) \geq c_\phi \cdot M$$

for a constant $c_\phi > 0$ bounded below independently of M : the witness bound (12) gives $\mathbb{E}[\lambda_{\max}] \geq (1 - \delta_h) q_h M_h \geq (1 - \delta_h) q_h \beta M$, where $\beta > 0$ is the width-ratio floor $M_h \geq \beta M$ supplied by the head hypothesis, so one may take $c_\phi := (1 - \delta_h) q_h \beta$. The witness route delivers this with no additive correction; within Karakida et al.’s Karakida et al. [2019] proportional-width model class, their Theorem 4 sharpens the width-linear rate to an exact asymptotic law. By Lemma 18,

$$\lambda_{\max}(G_{\theta\theta}) \leq \|J_\theta\|_{\text{op}}^2 \leq C_\theta$$

with C_θ independent of M and C_g . The cap is deterministic—valid for every realization of the initialization—so it divides through inside the expectation:

$$\mathbb{E}[\mathcal{D}_{\text{curv}}^{(M)}] = \mathbb{E}\left[\frac{\lambda_{\max}(G_{\phi\phi}^{(M)})}{\lambda_{\max}(G_{\theta\theta})}\right] \geq \frac{\mathbb{E}[\lambda_{\max}(G_{\phi\phi}^{(M)})]}{C_\theta} \geq \frac{c_\phi \cdot M}{C_\theta} = c_1 \cdot M, \quad (13)$$

where $c_1 := c_\phi/C_\theta$ depends only on the initialization scheme, the architecture class of g_ϕ , the classifier’s input Lipschitz constant $\tilde{L}$, the bottleneck dimension K , the input spectral dimension S , and the data distribution—exactly the dependencies claimed in the theorem statement. The witness route needs no additive correction ($c_2 = 0$ suffices); the subtracted constant is retained in the statement so that sharper asymptotic constants, which hold only for M large, may also be substituted. Under $C_g = \Theta(M^2)$, $M = \Theta(\sqrt{C_g})$ and the ratio form $\mathcal{D}_{\text{curv}} \geq c'_1 \sqrt{C_g/C_f} - c_2$ follows on multiplying and dividing by $\sqrt{C_f}$. This completes the proof. $\square$

Remark 42 (What Theorem 14 does and does not claim). Theorem 14 is a scaling law for the curvature ratio between two Gauss–Newton blocks; it does not claim that the full joint Gauss–Newton matrix has a large condition number in the strict $\lambda_{\max}/\lambda_{\min}$ sense (which is trivially infinite under rank-deficiency), nor that joint gradient descent produces disparate effective learning rates (which is the separate content of Theorem 27, under its own hypotheses). The scaling $\mathcal{D}_{\text{curv}} = \Omega(M)$ (equivalently $\Omega(\sqrt{C_g/C_f})$ for $\Theta(M^2)$ -parameter families) is a purely geometric statement about how curvature concentrates in the downstream module as the spatial network widens.

Remark 43 (Width scaling and the empirical exponent). The $\sqrt{C_g}$ scaling reflects the fact that Karakida’s underlying result Karakida et al. [2019] (Theorem 4) gives $\lambda_{\max}$ in terms of the network’s characteristic *width* M , not its total parameter count. For a fixed-depth architecture with $C_g = \Theta(M^2)$, the equivalent scaling is $\lambda_{\max}(J_\phi^\top J_\phi) \sim \sqrt{C_g}$ (log–log slope 0.5 against C_g). Experiment 1.1’s SpatialMLP is *not* in this class: its parameter count is linear in width ($C_g = \Theta(M)$), so the applicable prediction there is slope 1.0 against C_g , and the measured 0.70 ± 0.05 is sublinear (Remark 39). The two parameter-count forms must not be conflated; the theorem’s native variable is M .

A.2 Supplementary material for Theorem 27

The main-text proof of Theorem 27 is complete for the gradient-flow model (7): the displacement bound follows by direct integration of the residual envelope, with the operator-norm cap supplied by Lemma 18. The supplementary work concerns the hypotheses and the discrete extension.

Remark 44 (Open extensions). Three extensions are deliberately left open rather than sketched. (i) *Discrete SGD*: a Robbins–Monro tracking argument (Borkar Borkar [2008], Ch. 2) should carry the displacement bound to constant- and decaying-step SGD, with gradient noise entering the constants; we have not carried this out and state Theorem 27 for (almost-everywhere) gradient flow only. (ii) *Deriving the envelope*: obtaining Assumption 24—in particular the growth of μ with C_g —from the architecture, e.g. by an NTK analysis of the spatial pathway at large width adapting Pezeshki et al. Pezeshki et al. [2021], is the missing link that would chain Theorems 14 and 27 into a single causal statement; we test its content empirically (Experiment 1.2) instead of claiming it. (iii) *Envelope verification*: fitting $C/(1 + \mu t)$ envelopes to measured residual trajectories, with the fitted $\hat{\mu}$ increasing in capacity, is part of the experimental programme (Section 7).

A.3 Empirical robustness

We here report robustness experiments referenced in the main text.

A.3.1 Optimizer ablation (SGD vs Adam)

[TODO: Phase 5 follow-up. Show EGR collapse pattern in both regimes; note Adam masks the natural gradient ratio.]

A.3.2 Loss-variant ablation

[TODO: Phase 5 follow-up. Cross-entropy, focal loss, label smoothing.]

A.3.3 Regularizer ablation

[TODO: Phase 5 follow-up. Weight decay, dropout, batch norm. Pezeshki et al. Pezeshki et al. [2021], App. B, already show gradient starvation is robust to these; we reproduce briefly in our setting.]

A.3.4 Smallest capacity ratio where the pathology disappears

[TODO: Phase 5 follow-up. Sweep C_g/C_f from 1 to 1000, identify the threshold below which joint training reaches frozen performance.]